

Self-Built Simple Quadruped Robot Dog - Final Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Self-Built Simple Quadruped Robot Dog
Course	Microprocessor Controlled Systems
Document Type	Final report
Primary Role	Team lead, mechanical designer, and controls developer
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Executive Summary

This project built an eight-servo quadruped robot dog using a 3D-printed structure, ESP32 controller, PCA9685 servo driver, MATLAB gait reasoning, and Arduino implementation.

The main engineering value was system integration. The robot had to be mechanically serviceable, electrically reliable, and controllable enough to demonstrate multiple behaviors.

The final robot completed forward walking, standing, sitting, waving, hip lift, and rocking behaviors, earning a course grade of A+.

Applicant Contribution Signals

- Designed a repairable 3D-printed body around servo replacement, wire routing, battery placement, and center-of-gravity constraints.
- Connected MATLAB gait simulation and inverse-kinematics reasoning to embedded servo output.
- Calibrated servo direction, neutral angle, mechanical offsets, and gait timing through repeated testing.
- Led the team through failures that crossed mechanics, electronics, software, and assembly.

Mechanical Architecture

The body was designed as a working robot platform rather than a display model. Repair access, repeated testing, battery placement, and wire routing were part of the structure.

The design supported continuous iteration because servos and electronics could be accessed during debugging.

Control Implementation

MATLAB was used to reason about gait trajectories and reachable workspace before translating the logic into Arduino code.

The ESP32 and PCA9685 controlled eight MG90S servos. Real behavior required calibration of direction, neutral angle, offsets, and update timing.

Validation

The final demonstration showed multiple behaviors on one platform, proving that the mechanical geometry and embedded control software worked together.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Compact eight-servo body	Designed a serviceable 3D-printed structure	Made repair and calibration possible	Mechanical packaging
Simulation-hardware mismatch	Calibrated servo direction and offsets	Reduced model-to-robot mismatch	System debugging
Multiple behaviors	Separated gait update, IK, and servo output	Made behavior tuning easier	Embedded control
Team integration	Coordinated hardware and software fixes	Kept testing moving through failures	Leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.